1. PRIOR TO THE COMMENCEMENT OF WORK ON-SITE.

THE PLUMBING CONTRACTOR IS STRONGLY ADVISED TO ATTEND A MEETING WITH THE HK SOLUTIONS DESIGN AND CONSTRUCTION ADMINISTRATION TEAM TO DISCUSS THE PROJECT.
ANY CHANGES TO FIXTURES, FITTINGS, OR MATERIALS SPECIFIED MUST BE APPROVED BY THE SUPERINTENDENT, INCLUDING PUMPS. FAILURE TO NOTIFY WILL RESULT IN REMOVAL OF ITEMS AT THE CONTRACTOR'S EXPENSE. A MATERIAL CHANGE SUBMISSION FORM, INCLUDED IN THIS SPECIFICATION, MUST BE COMPLETED AND CLIENT-APPROVED PRIOR TO MAKING CHANGES. SUBMIT COMPLETED FORMS TO HK SOLUTIONS BEFORE FORWARDING TO THE CLIENT.
THE CONTRACTOR MUST KEEP THE BUILDER INFORMED OF COMPLETED WORKS TO ENSURE TIMELY HYDRAULIC INSPECTIONS.

2. LIMITATIONS OF DRAWINGS

DRAWINGS ARE NOT FOR INSTALLATION USE. "CONSTRUCTION" DRAWINGS ARE FOR PREPARING DETAILED SHOP DRAWINGS. THE PLUMBING CONTRACTOR MUST INSPECT THE SITE, GATHER MEASUREMENTS, AND MEET SITE-SPECIFIC NEEDS.
THE CONTRACTOR MUST ALSO ENSURE SHOP DRAWINGS ALIGN WITH THE LATEST ARCHITECTURAL, STRUCTURAL, AND SERVICES DRAWINGS DURING CONSTRUCTION.

3. AUTHORITIES & STANDARDS

ALL WORK MUST BE SUPERVISED BY A LICENSED PLUMBING CONTRACTOR, ADHERING TO THE DRAWINGS, SPECIFICATIONS, AND APPROVALS FROM THE SUPERINTENDENT AND RELEVANT AUTHORITIES.
WORK MUST MEET OR EXCEED CURRENT AUSTRALIAN STANDARDS. PROPOSALS FOR ALTERNATIVE ITEMS REQUIRE WRITTEN CONFIRMATION FROM THE MANUFACTURER VERIFYING COMPLIANCE WITH THE RELEVANT STANDARDS. APPROVAL MUST BE OBTAINED BEFORE ORDERING MATERIALS OR STARTING WORK.
IF STANDARDS OR DESIGN COMPLIANCE ARE UNCLEAR, THE SUPERINTENDENT MUST BE CONSULTED BEFORE WORK BEGINS. CLAIMS FOR REDUNDANT WORK WILL NOT BE CONSIDERED WITHOUT PRIOR NOTIFICATION.
THE PLUMBING CONTRACTOR IS RESPONSIBLE FOR LIAISING WITH AUTHORITIES TO MEET THEIR REQUIREMENTS. DESIGN, SUPPLY, AND INSTALL ALL PROJECT WORKS IN ACCORDANCE WITH RELEVANT STANDARDS AS SPECIFIED.

STANDARD	YEAR	DESCRIPTION
AS 2441	2005	INSTALLATION OF FIRE HOSE REELS
AS 2419.1	2021	FIRE HYDRANT INSTALLATIONS PART 1: SYSTEM DESIGN, INSTALLATIONSAND COMMISSIONING
AS 2444	2001	PORTABLE FIRE EXTINGUISHERS AND FIRE BLANKETS – SELECTION AND LOCATION
AS 2941	2013	FIXED FIRE PROTECTION INSTALLATION – PUMPSET SYSTEMS
AS 3500.0	2021	PLUMBING AND DRAINAGE – GLOSSARY OF TERMS
AS 3500.1	2021	PLUMBING AND DRAINAGE – WATER SERVICES
AS 3500.2	2021	PLUMBING AND DRAINAGE – SANITARY PLUMBING AND DRAINAGE
AS 3500.3	2021	PLUMBING AND DRAINAGE – STORMWATER DRAINAGE
AS 3500.4	2021	PLUMBING AND DRAINAGE – HEATER WATER SERVICES
AS 5601.1	2022	GAS INSTALLATIONS – PART 1: GENERAL INSTALLATIONS
AS 5601.2	2022	GAS INSTALLATIONS - PART 2: LP GAS INSTALLATIONS IN CARAVANS AND BOATS FOR NON PROPULSIVE PURPOSES
EP&A		STATE ENVIRONMENTAL PLANNING AND ASSESSMENT ACT
NCC VOLUME 1	2022	BUILDING CLASS 2 - 9
NCC VOLUME 2	2022	BUILDING CLASS 1 & 10
NCC VOLUME 3	2022	PLUMBING CODE OF AUSTRALIA

THE STANDARDS REFERENCED IN THE TABLE ABOVE ARE PRIMARY REFERENCES ONLY. REFER TO THE RELEVANT STANDARD FOR SECONDARY REFERENCES AS NECESSARY.

4. APPLICATIONS, FEES AND CHARGES

THE PLUMBING CONTRACTOR SHALL BE RESPONSIBLE FOR MAKING APPLICATION AND PAYING ALL FEES AND CHARGES RENDERED BY THE RESPECTIVE AUTHORITIES IN REFERENCE TO THE PROJECT.

5. CERTIFICATION OF WORKS

BEFORE FINAL PAYMENT, THE PLUMBING CONTRACTOR MUST APPLY FOR, PAY FEES, AND PROVIDE THE SUPERINTENDENT WITH CERTIFICATES CONFIRMING COMPLIANCE WITH CURRENT REGULATIONS AND AUTHORITY REQUIREMENTS.

IF THE RELEVANT AUTHORITY DOES NOT ISSUE A CERTIFICATE, THE CONTRACTOR MUST SUPPLY A CERTIFICATE OR LETTER OF CERTIFICATION GUARANTEEING COMPLIANCE WITH THE AUTHORITY'S REGULATIONS AND CONDITIONS.

6. DIMENSIONS AND LEVELS

INVERT LEVELS ON THE DRAWINGS ARE RECOMMENDATIONS ONLY AND MUST BE VERIFIED ON-SITE BEFORE EXCAVATION OR PIPE INSTALLATION TO ENSURE PROPER CONNECTIONS, COVER, AND FALL. REPORT DISCREPANCIES TO THE SUPERINTENDENT PRIOR TO STARTING WORK; CLAIMS FOR REDUNDANT WORK DUE TO NON-COMPLIANCE WILL NOT BE ACCEPTED.

CONFIRM THE DEPTH, POSITION, AND SUITABILITY OF SEWER AND STORMWATER CONNECTION POINTS BEFORE STARTING PIPE LAYING OR EXCAVATION, ENSURING PROPER GRAVITATION. NOTIFY THE SUPERINTENDENT IMMEDIATELY IF ADJUSTMENTS ARE NEEDED.

NO PIPE LAYING OR EXCAVATION IS PERMITTED UNTIL THESE STEPS ARE COMPLETED. CLAIMS FOR REDUNDANT WORK DUE TO NON-COMPLIANCE WILL NOT BE CONSIDERED.

7. TESTING

INCLUDE THE COST OF ALL REQUIRED TESTS AS SPECIFIED OR MANDATED BY AUTHORITIES. PROVIDE NECESSARY MATERIALS AND APPARATUS FOR TESTING. UNDERGROUND OR ENCLOSED WORK MUST REMAIN UNCOVERED UNTIL INSPECTED, TESTED, AND APPROVED BY THE SUPERINTENDENT AND RELEVANT AUTHORITIES.
UPON COMPLETION, CONDUCT PROCEDURES TO CONFIRM SYSTEM FUNCTIONALITY UNDER NORMAL CONDITIONS AS REQUESTED BY THE SUPERINTENDENT AND AUTHORITIES.
GIVE AT LEAST TWO (2) WORKING DAYS' NOTICE BEFORE TESTING BEGINS. KEEP UPDATED DRAWINGS ON-SITE TO TRACK TESTING PROGRESS, WITH EACH APPROVED SECTION INITIALLED BY THE SUPERINTENDENT. FIX ANY DEFECTS FOUND DURING TESTING AND RE-TEST AS REQUIRED.

8. FIXING & SUPPORTING OF PIPES

WHERE REQUIRED, AS PER THE SEISMIC RESTRAINT SECTION OF THE SPECIFICATION, ALL PIPEWORK SHALL BE PROVIDED WITH SEISMIC RESTRAINTS COMPLYING WITH AS1170.4, PROVIDING STABILISATION BOTH TRANSVERSELY AND LONGITUDINALLY. HANGER RODS SHALL ALSO BE PROVIDED WITH ROD STIFFENERS TO PREVENT BUCKLING DURING SEISMIC EVENTS. SUPPORTS AND FIXINGS TO AS1170.4 ARE TO BE DESIGNED BY SUITABLY QUALIFIED PERSONS AND BE SUITABLE FOR THE APPLICATION AND MATERIAL TO BE SUPPORTED.

ALL DIMENSIONS SHOWN IN MM. DO NOT SCALE FROM DRAWINGS. DIMENSIONS TAKE PRECEDENCE. IF IN DOUBT REFER TO HK SOLUTIONS.

9. SEISMIC RESTRAINT OF NON-STRUCTURAL COMPONENTS

ALL PIPEWORK SHALL BE PROVIDED WITH SEISMIC RESTRAINTS COMPLYING WITH AS1170.4, PROVIDING TRANSVERSELY AND LONGITUDINALLY STABILISATION. HANGER RODS SHALL ALSO BE PROVIDED WITH ROD STIFFENERS TO PREVENT BUCKLING DURING SEISMIC EVENTS.
DUCTILE MATERIAL SPACING
THE MAXIMUM ALLOWABLE BRACE SPACING FOR PIPING CONSTRUCTED OF DUCTILE MATERIALS (E.G. STEEL, COPPER, ALUMINIUM) ARE:
a. 12M FOR TRANSVERSE BRACING (PIPING LARGER THAN 65 MM DIA.);
b. 24M FOR LONGITUDINAL BRACING (PIPING LARGER THAN 65 MM DIA.);
c. 9M FOR TRANSVERSE BRACING (PIPING SMALLER THAN 65 MM DIA.);
d. 18M FOR LONGITUDINAL BRACING (PIPING SMALLER THAN 65 MM DIA.).
NON-DUCTILE MATERIAL SPACING
THE MAXIMUM ALLOWABLE BRACE SPACING LIMITS FOR PIPING NON-DUCTILE MATERIALS (E.G. CAST IRON, PLASTIC) ARE:
a. 6 M FOR TRANSVERSE BRACING;
b. 12 M FOR LONGITUDINAL BRACING.

10. CONCRETE

ALL CONCRETE USED IN CONJUNCTION WITH THIS WORK SHALL HAVE A MINIMUM STRENGTH OF 20 MPA AT TWENTY-EIGHT DAYS WHEN TESTED IN ACCORDANCE WITH AS 1012 PART 8 - METHOD FOR MAKING AND CURING CONCRETE COMPRESSION, INDIRECT TENSILE AND FLEXURE TEST SPECIMENS IN THE LABORATORY OR IN THE FIELD.

11. GREASE TRAP DECOMMISSIONING PROCESS

THE EXISTING WASTEWATER GREASE TRAPS ARE TO BE BYPASSED AND DECOMMISSIONED USING THE PROCEDURE DETAILED BELOW. THIS IS TO ENSURE THAT THE REDUNDANT TANK(S) DO NOT CAUSE ANY FUTURE PUBLIC HEALTH OR ENVIRONMENTAL PROBLEMS.

PROCEDURE
1. ALL EFFLUENT AND SLUDGE IS TO BE REMOVED FROM THE REDUNDANT GREASE TRAP(S) BY AN APPROVED CONTRACTOR. NOTE: RETAIN THE PUMPOUT RECEIPT FOR VERIFICATION.
2. AFTER PUMPOUT/CLEANOUT, TANK(S) WILL BE TREATED WITH AG LIME OR HYDRATED LIME TO DISINFECT AND NEUTRALISE THE TANKS. CARE SHOULD BE TAKEN WHEN USING LIME. IT IS RECOMMENDED THAT APPROPRIATE PRECAUTIONARY STEPS BE TAKEN SUCH AS USING PERSONAL PROTECTIVE EQUIPMENT (SKIN PROTECTION, EYE PROTECTION AND RESPIRATORY PROTECTION). A COPY OF THE MSDS FOR LIME SHOULD BE ON HAND. ONLY PROPERLY QUALIFIED PERSONS SHOULD BE EMPLOYED TO PERFORM ANY PROCEDURES USING CHEMICALS.
3. AFTER TREATMENT WITH LIME, A HOLE OR HOLES ADEQUATE FOR DRAINAGE PURPOSE ARE TO BE PROVIDED TO THE BOTTOM OF THE TANK (S).
4. THE CONCRETE LIDS AND PORTION OF THE TANK WALLS ARE TO BE BROKEN TO BELOW GROUND LEVEL.
5. THE TANK (S) IS TO BE FILLED WITH SOLID, NON-PUTRESCIBLE FILL; WITH THE GROUND SURFACE MADE GOOD. IF SETTLING OF THE FILL MATERIAL OCCURS OVER TIME IT MAY BE NECESSARY TO MAKE GOOD THE GROUND SURFACE WITH FURTHER FILL.

12. INSTALLATION CO-ORDINATION

VARIOUS APPARATUS AND EQUIPMENT WILL BE INSTALLED BY OTHERS. THE PLUMBING CONTRACTOR MUST REVIEW SUBCONTRACTOR PLANS, ENSURE COORDINATION, AND SCHEDULE WORK TO AVOID DELAYS.

SPACE REQUIREMENTS MUST BE CHECKED TO ENSURE EQUIPMENT AND PIPING FIT ALLOTTED SPACES. TYPICAL ARRANGEMENTS, SUCH AS CORE TOILETS, MUST BE COMPLETED AND APPROVED BEFORE DUPLICATION.

THE PLUMBING CONTRACTOR MUST COORDINATE INSTALLATIONS WITH OTHER SUBCONTRACTORS, ADDRESSING FITTING CLASHES WITHIN THE TENDER PRICE. DIMENSIONS MUST BE VERIFIED ON-SITE PRIOR TO STARTING WORK.

RESPONSIBILITY FOR COORDINATION AND BUILDABILITY REMAINS WITH THE PLUMBING CONTRACTOR, INCLUDING INTEGRATING POST-CONTRACT DESIGN CHANGES FROM OTHER TRADES.

ALL SERVICE ELEMENTS OF THIS SUBCONTRACT MUST BE INSTALLED WITHIN DESIGNATED SPACES, COORDINATING WITH ALL OTHER TRADE DESIGNS.

13. INSPECTION & FLUSHING OUT OF SERVICES

BEFORE APPLYING FOR THE CERTIFICATE OF PRACTICAL COMPLETION, INSPECT GRAVITY LINES BY LIFTING ACCESS CHAMBER COVERS AND GATES, CLEARING DEBRIS, FLUSHING THE SYSTEM, AND REPLACING COVERS AFTER COMPLETION. NOTIFY THE SUPERINTENDENT AT LEAST 48 HOURS BEFORE INSPECTION AND FLUSHING.

FOR PRESSURE SYSTEMS, FLUSH PIPING TO REMOVE DEBRIS LIKE SAND AND FILINGS. DISCONNECT ANY COMPONENTS VULNERABLE TO DAMAGE, SUCH AS VALVES, FILTERS, AND BACKFLOW PREVENTERS. ENSURE THE SYSTEM IS FULLY CLEAR, REPLACE DAMAGED EQUIPMENT, AND RECONNECT AS NEEDED. PROVIDE 48 HOURS' NOTICE TO THE SUPERINTENDENT BEFORE THESE PROCEDURES.

14. FOOD PREMISES

ALL WORK WITHIN THE KITCHEN / FOOD AREAS SHALL BE IN ACCORDANCE WITH THE RELEVANT SECTIONS OF AS4674 DESIGN, CONSTRUCTION AND FIT-OUT OF FOOD PREMISES, THE BUILDING CODE OF AUSTRALIA (BCA), THE LOCAL COUNCIL GUIDELINES/REQUIREMENTS AND THE STATES HEALTH DEPARTMENT.

15. TENDER DOCUMENTS

THE DRAWINGS, AS SCHEDULED, ARE ISSUED AS A GUIDE ONLY AND SHALL BE CONSIDERED TO BE DIAGRAMMATIC AND APPROXIMATE. THE DRAWINGS AND SPECIFICATIONS ARE INTENDED TO BE MUTUALLY EXPLANATORY AND COMPLETE, BUT ALL WORK CALLED FOR BY ONE, EVEN IF NOT BY THE OTHER, SHALL BE FULLY EXECUTED. SHOULD THE DOCUMENTS BE IN CONFLICT, THE PLUMBING CONTRACTOR SHALL BE DEEMED TO HAVE INCLUDED THE LARGER QUANTITY AND/OR THE MORE EXPENSIVE COMPONENTS, AS APPLICABLE.

SHOULD THE DRAWINGS AND SPECIFICATIONS INDICATE AMBIGUITIES THE REQUIREMENTS OF THE SPECIFICATION SHALL TAKE PRIORITY. UNLESS THE PLUMBING CONTRACTOR ADVISES REGARDING AMBIGUITIES OR CONFLICTS AT THE TIME OF TENDER, THE REQUIREMENTS OF THE SPECIFICATION WILL BE EXPECTED AS FULFILMENT OF THE PLUMBING CONTRACT.

16. DOCUMENTS BEFORE PRACTICAL COMPLETION

BEFORE APPLYING FOR THE CERTIFICATE OF PRACTICAL COMPLETION, A SET OF "AS BUILT" DRAWINGS (COMPRISING ALL FLOOR PLANS, DIAGRAMMATICS AND DETAILS) PREPARED BY THE PLUMBING CONTRACTOR TO A MINIMUM SCALE GREATER THAN OR EQUAL TO THAT OF THE CONTRACT DRAWINGS AND RELEVANT BOUND MANUALS INDICATING ALL ITEMS OF SERVICE OR EQUIPMENT SHALL BE SUBMITTED TO THE SUPERINTENDENT FOR REVIEW. THE STANDARD OF DRAFTING AND MINIMUM SCALES SHALL BE EQUAL TO DESIGN DOCUMENTS, ACCURATELY PLOTTED, SHOWING DIMENSIONS OF PIPES, VALVES AND SERVICES TO PROVIDE:
1. AS-BUILT DRAWINGS IN PDF AND AUTOCAD / REVIT FORMATS (MATCHING TENDER DOCUMENTATION FORMAT)
2. OPERATION AND MAINTENANCE MANUALS
3. UPDATED SEWERAGE SERVICES DIAGRAM FOR LOCAL AUTHORITY
4. DETAILED CERTIFICATES OF COMPLIANCE.
5. TEST INFORMATION AND CERTIFICATES FOR EACH
6. THERMOSTATIC MIXING VALVE
7. BACKFLOW PREVENTION DEVICE
6 FLOW AND PRESSURE TEST INCLUDING TEST INFORMATION DATA
a. FIRE HYDRANT SYSTEM
b. FIRE HOSE REEL SYSTEM
c. DRENCHERS WALL WETTING SPRINKLERS
d. FIRE SEALS, FIRE COLLARS AND SEALING OF ALL PENETRATIONS.
7. ACCESS COVER LIFTING KEYS
8 HOSE TAP OPERATING KEYS

17. COMPACTION SCHEDULE

BEFORE PRACTICAL COMPLETION, PROVIDE:
1. AS-BUILT DRAWINGS IN PDF AND AUTOCAD / REVIT FORMATS (MATCHING TENDER DOCUMENTATION FORMAT)
2. OPERATION AND MAINTENANCE MANUALS
3. UPDATED SEWERAGE SERVICES DIAGRAM FOR LOCAL AUTHORITY
4. DETAILED CERTIFICATES OF COMPLIANCE.
5. TEST INFORMATION AND CERTIFICATES FOR EACH
6. THERMOSTATIC MIXING VALVE
7. BACKFLOW PREVENTION DEVICE
6 FLOW AND PRESSURE TEST INCLUDING TEST INFORMATION DATA
a. FIRE HYDRANT SYSTEM
b. FIRE HOSE REEL SYSTEM
c. DRENCHERS WALL WETTING SPRINKLERS
d. FIRE SEALS, FIRE COLLARS AND SEALING OF ALL PENETRATIONS.
7. ACCESS COVER LIFTING KEYS
8 HOSE TAP OPERATING KEYS

18. SUPPLY OF MATERIALS

SUPPLY AND FIX ALL MATERIALS REQUIRED TO COMPLETE THE WORK. ALL MATERIALS SHALL BE FIRST QUALITY AND THE BEST OF THEIR RESPECTIVE KINDS. SECOND-QUALITY OR INFERIOR MATERIALS SHALL BE REJECTED. ALL COSTS ASSOCIATED WITH THE REPLACEMENT OF REJECTED MATERIALS SHALL BE BORNE BY THE PLUMBING CONTRACTOR. ALL MATERIALS SHALL CONFORM TO THE LATEST AUSTRALIAN STANDARD SPECIFICATION, CODE OR INTERIM CODE. IF NO AUSTRALIAN STANDARD EXISTS, THEY SHALL CONFORM TO THE LATEST BRITISH STANDARD OR THE AMERICAN SOCIETY FOR TESTING AND MATERIALS IN THAT ORDER.

19. FIRE STOP COLLARS & PENETRATIONS

ALLOW TO SUPPLY AND INSTALL APPROVED FIRE STOP COLLARS TO ALL PIPE PENETRATIONS THROUGH SLABS, WALLS AND ALL FIRE SEPARATED AREAS.
FIRE STOP COLLARS SHALL BE PROVIDED ON ALL PVC-U, HDPE AND NON-METAL PIPE PENETRATIONS THROUGH FIRE RATED REINFORCED CONCRETE SLABS.
ALL PENETRATIONS AND CORE HOLES TO BE PROTECTED IN ACCORDANCE WITH NCC 2022 CLAUSE C4D15.
FIRE COLLARS TO BE INDICATED ON SHOP DRAWINGS AND FIRE COLLAR MATRIX TO BE PROVIDED.
ALL FIRE COLLARS TO HAVE ASSOCIATED TEST REPORTS PROVIDED.

20. TRENCH EXCAVATION

KEEP ADEQUATE DEWATERING EQUIPMENT ON THE SITE AT ALL TIMES AND KEEP EXCAVATION FREE OF WATER.
ALL EXCAVATION SHALL BE IN ACCORDANCE WITH WORKCOVER - CODE OF PRACTICE PER STATE FOR EXCAVATION.
ALL EXCAVATIONS SHALL BE CARRIED OUT IN A SAFE, SECURE AND PRUDENT MANNER THAT WILL PREVENT ANY FORM OF COLLAPSE, SLIP OR FAILURE OF THE SIDES OF THE EXCAVATION.
THE GROUND SHALL BE EXCAVATED IN THE FORM OF TRENCHES TO ENABLE THE VARIOUS PIPELINES TO BE CONSTRUCTED IN THE LOCATIONS SHOWN ON THE DRAWINGS WITH TRENCH WIDTH INCREASED WHERE NECESSARY TO PERMIT THE CONSTRUCTION OF ACCESS CHAMBERS AND PITS. TRENCHES SHALL BE EXCAVATED AT UNIFORM GRADES AND IN STRAIGHT LINES.
UPON COMMENCEMENT OF EXCAVATIONS A GENERAL EXAMINATION OF THE TRENCHES SHALL BE MADE TO DETERMINE THE PRESENCE OR OTHERWISE OF ANY MATERIAL CONSIDERED BY THE PLUMBING CONTRACTOR TO BE DETRIMENTAL TO THE LIFE AND FUTURE CONDITION OF THE PIPELINE TO BE INSTALLED.
WHEREVER SUCH MATERIAL IS LOCATED, THE PLUMBING CONTRACTOR SHALL ADVISE THE SUPERINTENDENT IMMEDIATELY TOGETHER WITH A RECOMMENDATION AND COSTS OF REMEDIAL ACTION TO BE TAKEN. NO WORK IN REGARD TO THIS REMEDIAL PRECAUTION IS TO COMMENCE UNTIL WRITTEN APPROVAL TO PROCEED IS OBTAINED FROM THE SUPERINTENDENT.
REMOVE ALL EXCESS EXCAVATION MATERIAL FROM SITE, PAY ALL CARTAGE AND DISPOSAL FEES WHEREVER APPLICABLE TO PROVIDE A CLEAN NEAT SITE DEVOID OF ALL DEBRIS CREATED BY THIS SECTION OF WORKS.

21. EXCEEDING EXCAVATION

IF THE PLUMBING CONTRACTOR HAS EXCEEDED THE SECTION AREA OF EXCAVATION, AS SHOWN, IN CONSEQUENCE OF ANY INJUDICIOUS WORKING, SLIPS, FALLS, OR FOR ANY CAUSE OTHER THAN BY DIRECTION OF THE SUPERINTENDENT THEN THE PLUMBING CONTRACTOR SHALL, AT SOLE COST, REMOVE SUCH EXTRA MATERIALS FROM THE SITE, MAKE GOOD AND FILL IN THE EXTRA EXCAVATION WITH CONCRETE, SAND OR APPROVED FILLING AS MAY BE DIRECTED.
NO EXTRA PAYMENT WILL BE MADE FOR EXCAVATION IN EXCESS OF THAT REQUIRED BY THE DRAWINGS AND SPECIFICATIONS UNLESS DIRECTED IN WRITING BY THE SUPERINTENDENT.

22. PRECAUTIONS & SAFEGUARDS

CARRY OUT THE WORK IN A CAREFUL, SECURE, SAFE AND TIDY MANNER AND TAKE ALL PRECAUTIONS AGAINST DAMAGE WHETHER ARISING FROM BAD WORKMANSHIP, BREAKAGE OR MACHINERY OR PLANT, INEFFICIENT TIMBERING, FLOODING OR ANY OTHER CAUSE WHATSOEVER. PROVIDE, ERECT AND MAINTAIN WARNING SIGNS, TEMPORARY FENCES, BARRIERS AND WARNING LIGHTS ADJACENT TO ANY WORKS SUCH AS TRENCHES AND EXCAVATIONS OR STACKS OF MATERIAL WHICH COULD BE CONSIDERED A DANGER TO PERSONS OR TRAFFIC OF ANY KIND.
THE PLUMBING CONTRACTOR SHALL OBEY ALL DIRECTIONS GIVEN TO HIM WITH REGARD TO THE PROVISIONS OF LIGHTING AND BARRIERS, BUT SHALL NOT BE RELIEVED OF THE RESPONSIBILITY TO PROVIDE ADEQUATE LIGHTING AND BARRIERS TO PREVENT ANY ACCIDENT OR DAMAGE.

23. BEDDING MATERIAL

WHERE TRENCH BASE IS OF STABLE SAND MATERIAL PIPE BARRELS SHALL BE DIRECTLY SUPPORTED ON THE UNDISTURBED NATURAL GROUND. IN ALL OTHER CASES, PIPELINES SHALL BE SUPPORTED ON EITHER A GRANULAR BEDDING OR CEMENT MORTAR BEDDING IN ACCORDANCE WITH AS/NZS 3500.2 CLAUSE 5.4 AND THE PLUMBING CODE OF AUSTRALIA.
WHERE COMPACTION OF BACKFILL OVER THE PIPELINE IS TO BE MADE BY THE USE OF MECHANICAL VIBRATOR, A FURTHER SPARGING TO A LEVEL EQUAL TO 1/3RD THE PIPE BARREL SHALL BE MADE USING 6:1 CEMENT MORTAR. THE ADDITIONAL CEMENT MORTAR SHALL BE PLACED AFTER THE INITIAL SETTING OF THE 4:1 SPARGING TAKES PLACE.
WHERE A MECHANICAL VIBRATOR IS EMPLOYED, THE LINES SHALL BE SUBJECTED TO A SECOND TEST UPON COMPLETION OF THE COMPACTION.

24. TRENCHES EXCAVATED IN ROCK

IT SHALL BE DEEMED THAT EXCAVATION IN ROCK HAS BEEN ALLOWED FOR IN THE TENDER WHERE IT SHOULD HAVE BEEN REASONABLY ANTICIPATED BY THE PLUMBING CONTRACTOR, OR WHERE TEST BORE RESULTS INDICATE THE PRESENCE OF ROCK AND THE TEST BORE RESULTS ARE AVAILABLE FOR INSPECTION BY THE PLUMBING CONTRACTOR, OR WHERE INSPECTION OF THE SITE AND ITS SURROUNDINGS WOULD INDICATE THE PRESENCE OF ROCK.
SHOULD THE PLUMBING CONTRACTOR ENCOUNTER ROCK NOT PREVIOUSLY ANTICIPATED DURING EXCAVATIONS, NOTIFY THE SUPERINTENDENT IMMEDIATELY SO THAT THE EXTENT, QUANTITY AND TYPE OF ROCK EXCAVATION CAN BE ASSESSED BY THE SUPERINTENDENT PRIOR TO COMMENCEMENT OF ROCK EXCAVATION. FAILURE TO OBSERVE THIS REQUIREMENT SHALL FORFEIT THE PLUMBING CONTRACTOR'S RIGHT TO REIMBURSEMENT OF ADDITIONAL COST'S RELATING TO THE EXCAVATION OF ROCK UNLESS THE EXTENT, QUANTITY AND TYPE OF ROCK CAN BE ASSESSED LATER AND THERE WAS NO ALTERNATIVE TO THE EXCAVATED ROUTE.
PROVIDE IN THE TENDER A SCHEDULE OF RATES, WHICH WILL BE THE BASIS OF ASSESSING THE COST OF ADDITIONAL WORK ASSOCIATED WITH EXCAVATION IN ROCK. THE SCHEDULE SHALL BE THE PLUMBING CONTRACTOR'S PRICE IN CUBIC METRE RATES FOR ADDITIONAL COST (EXTRA OVER) TO EXCAVATE TRENCHES IN SOFT ROCK AND IN HARD ROCK. THE PLUMBING CONTRACTOR SHALL ALSO PROVIDE CUBIC METRE RATES FOR THE ADDITIONAL COST TO EXCAVATE PITS IN SOFT ROCK AND IN HARD ROCK.

25. TRADE WASTE APPLICATION

IT IS THE RESPONSIBILITY OF THE PLUMBING CONTRACTOR TO MAKE THE TRADE WASTE APPLICATION WITH LOCAL AUTHORITY (OR THE LOCAL TRADE WASTE AUTHORITY) FOR THE INSTALLATION OF THE GREASE ARRESTOR
COORDINATE THE APPLICATION WITH THE CLIENT/BUILDING OWNER AND THE TRADE WASTE INSPECTOR.

26. WATER MAIN CONNECTION

MAKE AN APPLICATION TO THE WATER AUTHORITY, EXCAVATE IT, AND CONNECT IT TO THE AUTHORITIES MM DIAMETER WATER MAIN FOR THE FOLLOWING SERVICES AND EXTEND TO THE RETICULATION AS INDICATED ON THE DRAWINGS. THE WATER AUTHORITY SHOULD BE CONTACTED, AND AN APPLICATION MADE TO ARRANGE FOR A QUOTE FOR THE CONNECTION TO THE WATER MAIN. ALLOW TO PAY ALL FEES FOR THIS WORK.

27. LP GAS SERVICE

SUPPLY AND INSTALL ALL GAS SERVICE PIPES FROM THE EXISTING LP SYSTEM TO ALL FIXTURES AND APPLIANCES REQUIRING GAS. INCLUDE FOR ALL PIPEWORK, BENDS, OFFSETS, BRACKETS, VALVES AND SUNDRY EQUIPMENT REQUIRED FOR THE INSTALLATION. THE GAS INSTALLATION SHALL BE INSTALLED IN ACCORDANCE WITH AS/NZS 5601.1. ALL WORK CARRIED OUT TO INSTALL OR REPLACE ALL OR ANY PART OF NATURAL GAS SYSTEM CONTAINING MULTILAYER PIPE MUST BE IN ACCORDANCE WITH AS/NZS 5601.1 (2022) (GAS INSTALLATIONS) CLAUSE 5.2.11
WHERE MULTILAYER PIPE IS USED AS PART OF CONSUMER PIPING, THE FOLLOWING REQUIREMENTS FOR FIRE EMERGENCY ISOLATION SHALL APPLY:
- THE INSTALLATION SHALL BE FITTED WITH A SYSTEM THAT WILL SHUT OFF THE GAS SUPPLY WHEN THE FIRE SAFETY SYSTEM OPERATES.
- THE METHOD OF ISOLATION SHALL BE A SINGLE CLASS 1 SAFETY SHUT-OFF VALVE THAT MEETS THE REQUIREMENTS OF AS4629 AND AUTOMATICALLY ISOLATES THE GAS FLOW TO THE BUILDING WHEN DE-ENERGIZED. THIS VALVE SHALL

a. OPERATE ONLY WHEN AN ACTIVE FIRE SAFETY SYSTEM OPERATES; AND
b. BE LOCATED PRIOR TO ANY MULTI-LAYERED PIPE AND BE AS CLOSE AS PRACTICABLE TO THE GAS SUPPLY POINT TO THE BUILDING, AND READILY ACCESSIBLE.
c. THE SYSTEM SHALL PROVIDE PRESSURE PROVING OF THE DOWNSTREAM INSTALLATION PRIOR TO RESTORATION OF THE GAS SUPPLY.

ALTERNATIVELY, FOR A CLASS 1A BUILDING IN AUSTRALIA, OR DETACHED OR MULTI-UNIT DWELLING IN NEW ZEALAND, A SYSTEM OR DEVICE THAT AUTOMATICALLY SHUTS OFF THE GAS SUPPLY IF GAS TIGHTNESS IS ADVERSELY AFFECTED BY A FIRE EMERGENCY EVENT SHALL BE PROVIDED. THE SYSTEM OR DEVICE WHICH SHUTS OFF THE GAS SUPPLY SHALL BE INSTALLED IN A READILY ACCESSIBLE LOCATION AS CLOSE AS PRACTICABLE TO THE POINT OF SUPPLY AND PRIOR TO ANY MULTI-LAYER PIPE. THE DEVICE SHALL BE SIZED AND COMMISSIONED IN ACCORDANCE WITH MANUFACTURER NUISANCE TRIPPING. FURTHER INFORMATION CAN BE FOUND AT THE DEPARTMENT OF FAIR TRADING. WHERE THE OPERATING PRESSURE OF THE CONSUMER PIPING EXCEEDS 7KPA - ABOVE-GROUND CONSUMER PIPING SHALL BE IDENTIFIED WITH STICKERS. THE IDENTIFICATION MARKINGS SHALL BE PLACED

A. AT SPACINGS OF NOT MORE THAN 8M;
B. ADJACENT TO BRANCHES, JUNCTIONS, VALVES, WALL AND FLOOR PENETRATIONS;
C. AND ON ALL TAILPIPES.

THE IDENTIFICATION SHALL COMPLY WITH AS 1345 IN ALL OTHER RESPECTS.
WHERE CONSUMER PIPING IS INTENDED TO OPERATE AT A PRESSURE EXCEEDING 7KPA, OR FOR LP GAS 14KPA, THE TECHNICAL REGULATOR MAY REQUIRE PLANS AND SPECIFICATIONS TO BE SUBMITTED PRIOR TO THE COMMENCEMENT OF WORK ON THE INSTALLATION.
IF THE MAXIMUM OVER-PRESSURE EXCEEDS 7KPA FOR NATURAL GAS SUPPLIES OR 14KPA FOR LP GAS SUPPLIES THE MAXIMUM OVER-PRESSURE IS REQUIRED TO BE CLEARLY MARKED ON OR ADJACENT TO THE GAS SERVICE PRESSURE REGULATOR BY THE GAS SUPPLIER/GMS OWNER. IT WILL BE NECESSARY TO DETERMINE FROM THE NETWORK OPERATOR THE MAXIMUM PRESSURE THAT COULD OCCUR UNDER GAS PRESSURE REGULATOR FAULT CONDITIONS AT THE OUTLET OF THE CONSUMER BILLING METER OR THE OUTLET OF THE CYLINDER OR TANK REGULATOR.

TENDER ISSUE

REV.	DESCRIPTION	DATE	INIT.	COMPANY
T1	ISSUE FOR TENDER	17.06.2026	MC	HKS

REVISIONS

HK Solutions
ENGINEERING
PROJECT MANAGEMENT
DARWIN. GEELONG.
CAIRNS. TOWNSVILLE.
PERTH. **BALLARAT**
admin@hksolutions.com.au
www.hksolutions.com.au

Conditions of use. This document may only be used by Harris Kmon Solutions clients for the purpose for which it was prepared and must not be used by any other person or for any other purpose.

DRAWN	MC
CHECKED	AB
DESIGNED	MC
CHECKED	AB
APPROVED	
Date Approved	

THIS DRAWING MUST NOT BE USED FOR CONSTRUCTION UNLESS SIGNED AS APPROVED

CITY OF ROCKHAMPTON
ROCKHAMPTON SPORTS PRECINCT
554-700 YAAMBA ROAD NORMAN GARDENS

HYDRAULIC SERVICES SPECIFICATION

PAGE No	PROJECT No	DRAWING No	DISCIPLINE	REV.	SHEET SIZE
2 OF 4	25015 - H002		HYDRAULIC	T1	A1